CASE STUDY · PYTHON · SQL · MACHINE LEARNING

Telecom Customer Churn Analysis & Retention Strategy

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Summary

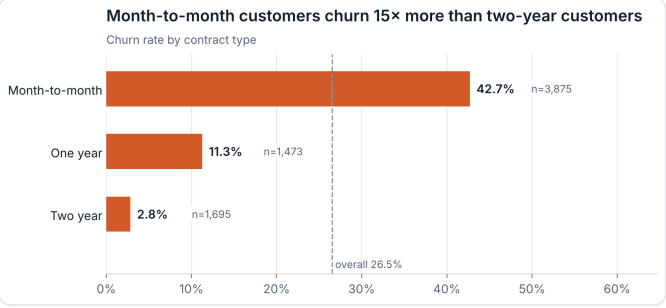
A telecom provider loses about a quarter of its customers, and nearly a third of its monthly revenue goes with them. I analysed 7,043 customer records to find who is leaving and why, measured the revenue at stake, and built a model that ranks every customer by churn risk so a retention team can act early.

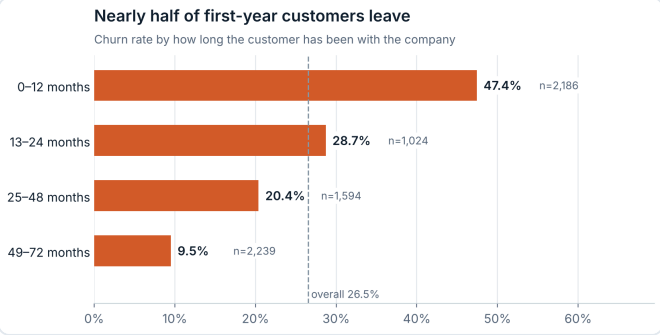
26.5%	$139K	0.845	50%
customers churned	monthly revenue lost (30.5%)	model ROC AUC on unseen data	of churners in the riskiest 20%

Approach

- **Data preparation (pandas):** checked duplicates and missing values; converted text totals to numbers (11 new customers with no bill yet were set to $0); added a churn flag and tenure bands; exported a clean file for Power BI.
- **SQL validation:** re-ran the key numbers as SQL queries in SQLite. The results matched pandas exactly.
- **Exploration:** compared churn rates across contract, tenure, internet service, payment method and support add-ons.
- **Modelling (scikit-learn):** logistic regression on a 75/25 stratified split, benchmarked against a random forest, and interpreted with odds ratios. I scored every customer with 5-fold cross-validation.

Key findings

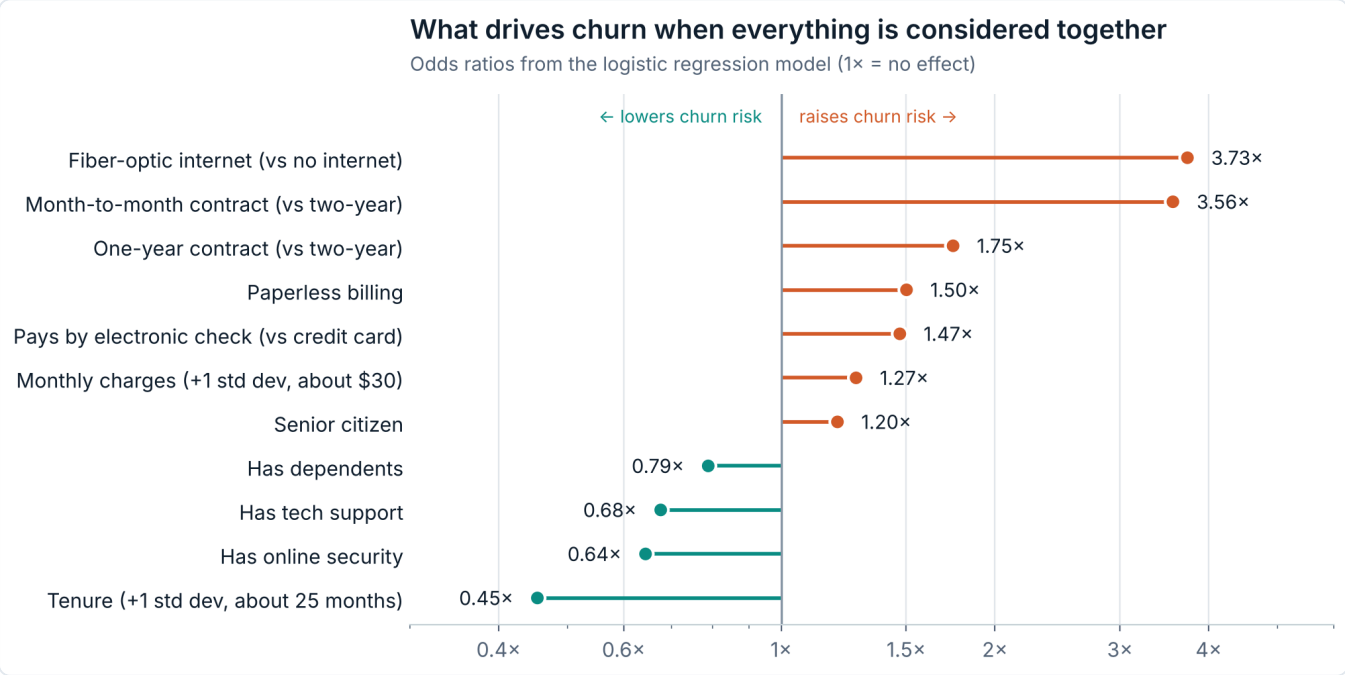


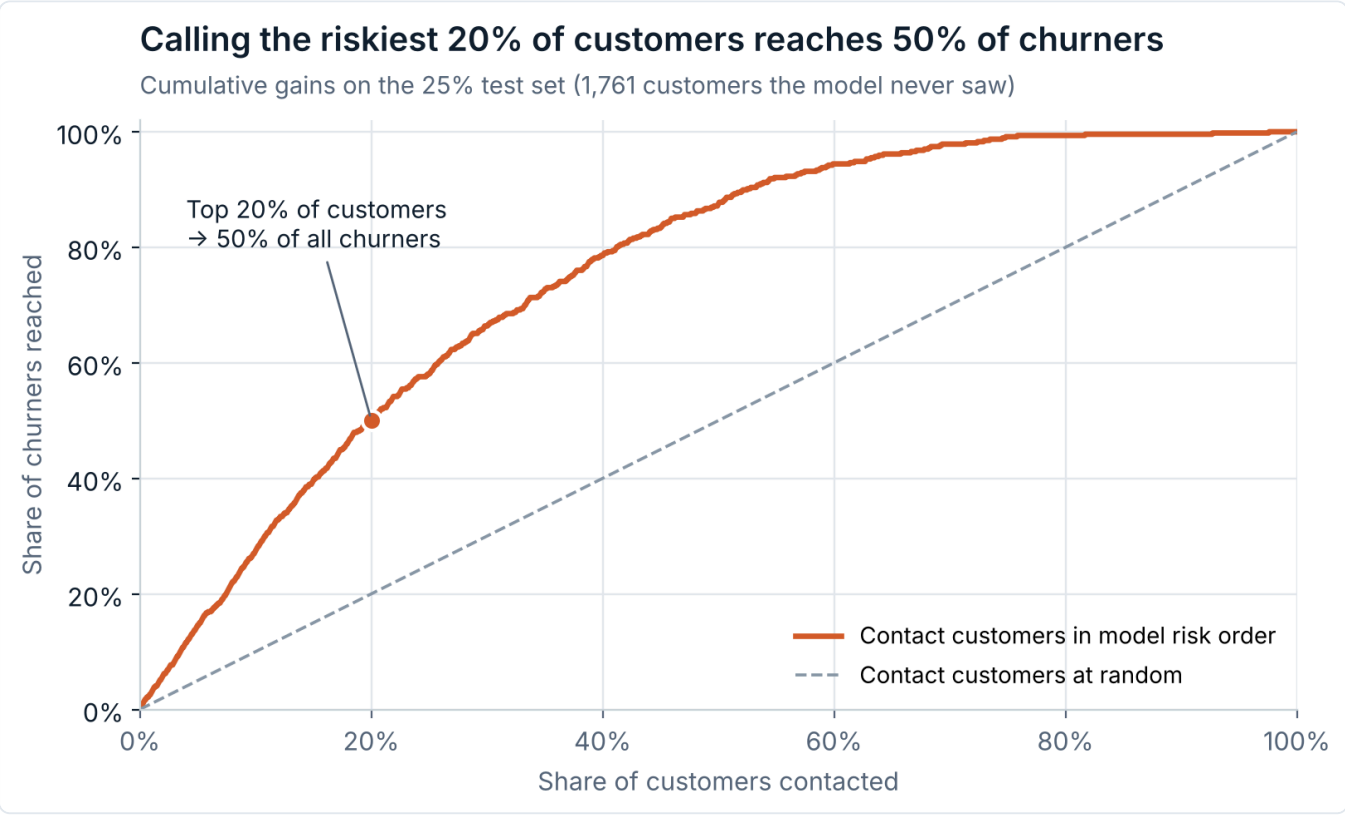


- Month-to-month customers churn at **42.7%** versus **2.8%** on two-year contracts, and they account for **86.9%** of lost revenue.
- **47.4%** of customers in their first year leave; the median churner had been with the company 10 months.
- Fiber-optic customers churn at **41.9%** despite paying the most; electronic-check payers churn at **45.3%**.
- Fiber customers without tech support churn at **49.4%**, versus **22.6%** with it.

Churn risk model

The logistic regression scored a ROC AUC of **0.845** on 1,761 customers it never saw, matching the more complex random forest (0.844). I kept it because each factor can be explained as an odds ratio.





Contacting the riskiest **20%** of customers reaches **50%** of all churners. In this high-risk tier, **67.6%** of customers actually left, and the tier holds **$115,981** of monthly revenue.

Recommendations

Action	Why
Move month-to-month customers onto contracts with a discount or perk, starting with high-risk customers	42.7% vs 2.8% churn; 3.6× the odds even after other factors are accounted for
Structured first-year onboarding (welcome call, check-ins at months 1, 3, 6)	47.4% churn in the first 12 months
Investigate the fiber-optic service: pricing, reliability, competitor offers	41.9% churn; 3.7× the odds of customers without internet
Bundle tech support and online security as a free trial, and A/B test it	49.4% vs 22.6% churn for fiber customers; odds 0.68× and 0.64×
Move electronic-check payers to auto-pay with a small incentive	45.3% vs 15–17% churn
Run the risk model monthly and give the retention team the high-risk list	Top 20% of customers contain 50% of churners

Limitations

- IBM sample data for a fictional company: the patterns are realistic, but the figures are not benchmarks.
- A single month's snapshot, with no history or seasonality.
- The findings show associations, not causes. Retention offers should be tested before full rollout.
- No cost data, so the return on each action cannot yet be calculated.

Skills demonstrated

Data cleaning and validation in pandas · SQL aggregation and CASE logic · exploratory analysis · data visualisation · logistic regression and model evaluation (AUC, precision/recall, cross-validation, gains curves) · turning analysis into business recommendations.

Data: IBM Telco Customer Churn sample dataset (github.com/IBM/telco-customer-churn-on-icp4d, Apache License 2.0). Full code: notebooks/telecom_churn_analysis.ipynb.